

CS-550 PA-1
Fatima Mariyam – A20527616
Output Document

3. Measuring how the transfer speed changes when varying the number of concurrent clients (N). All the clients are requested to download files from the server concurrently. Each client should create at least 10 threads downing a fixed number (set by students) of files simultaneously. and each client needs to repeat the experiments for a fixed number (set by students).

- a. N can be 2, 4, 8, 16
- b. Graph your results

Output:

Following are the durations in ns for N number of clients running concurrently, each with 10 threads and downloading a file of size 8kb.

Execute script run_experiment_3.sh with argument = 2(number of clients N)

➔ **./run_experiment_3.sh 2**

Running experiment with 2 clients...

Client created!

Client created!

Experiment durations:8082.88, 1982.75, 1454.88, 1332.50, 1211.25

Experiment durations:7701.13, 1812.83, 1381.54, 1266.46, 1429.17

Experiment with 2 clients completed.

Execute script run_experiment_3.sh with argument = 4(number of clients N)

➔ **./run_experiment_3.sh 4**

Running experiment with 4 clients...

Client created!

Client created!

Client created!

Client created!

Experiment durations:16449.63, 4446.42, 3980.75, 3412.08, 3298.21

Experiment durations:18087.54, 3475.92, 3582.54, 3734.08, 3125.17

Experiment durations:18197.42, 2822.00, 3054.17, 4545.17, 2885.75

Experiment durations:17536.04, 3806.25, 3594.25, 5137.88, 2942.63

Experiment with 4 clients completed.

Execute script run_experiment_3.sh with argument = 8(number of clients N)

➔ **./run_experiment_3.sh 8**

Running experiment with 8 clients...

Client created!

Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Experiment durations:23779.67, 4189.29, 3039.04, 4407.21, 3094.46
Experiment durations:23729.29, 4155.00, 4137.96, 3689.58, 3119.83
Experiment durations:21149.46, 6747.83, 3957.08, 3903.13, 3117.21
Experiment durations:25596.92, 4443.13, 3182.00, 4325.21, 3916.88
Experiment durations:7043.04, 1615.13, 1121.25, 1047.08, 1112.42
Experiment durations:8713.54, 1767.54, 3667.71, 3476.29, 4717.92
Experiment durations:7990.63, 4087.79, 2837.96, 2290.63, 1984.08
Experiment durations:9888.58, 2576.79, 1999.54, 1436.92, 2211.08
Experiment with 8 clients completed.

Execute script run_experiment_3.sh with argument = 16(number of clients N)

➔ **./run_experiment_3.sh 16**

Running experiment with 16 clients...

Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Client created!
Experiment durations:14771.13, 4186.50, 6855.29, 4837.88, 4236.08
Client created!
Client created!
Client created!
Client created!
Client created!
Experiment durations:8700.88, 1722.29, 1486.29, 1092.08, 1255.63
Experiment durations:8308.63, 1678.29, 1151.38, 1428.96, 1916.04
Experiment durations:9742.33, 2601.00, 2592.25, 1274.42, 2831.04
Experiment durations:12327.63, 4330.79, 5011.33, 4434.58, 136359.79
Experiment durations:11965.04, 4716.33, 4836.67, 4494.67, 128977.79
Experiment durations:8578.21, 2469.04, 1867.04, 3515.08, 3440.63
Experiment durations:10504.33, 2831.38, 1371.75, 2726.17, 8210.42

Experiment durations:10042.50, 2167.33, 2051.08, 4068.92, 4497.75
Experiment durations:11172.96, 4401.54, 174770.79, 10125.38, 40889.33
Experiment durations:221564.58, 50093.04, 21668.08, 3800.17, 5018.63
Experiment durations:231770.25, 49555.29, 21206.04, 6233.75, 4591.13
Experiment durations:61176.25, 9217.13, 6233.21, 5448.42, 2116.42
Experiment durations:10046.50, 1696.04, 1937.46, 19979.25, 6512.50
Experiment durations:10959.54, 2485.79, 1694.17, 9172.46, 18938.96
Experiment durations:14556.42, 3144.17, 141970.46, 10288.50, 3819.42
Experiment with 16 clients completed.

4. Measuring how the transfer speed changes when varying the download file size.
 - a. Different file sizes are 128, 512, 2k, 8k, 32k
 - b. For each size of file in a), performing the following test
 - 4 clients concurrently request to download files from the server. Each client creates at least 10 threads to download a fixed number (set by students) of files simultaneously, and each client needs to repeat the experiments for a fixed number (set by students).
 - c. Graph your results

Output:

Execution results for the above experiment with 4 clients, each with 10 threads downloading files of different given sizes.

Execute script run_experiment_4.sh

Running experiment for file size: **128b**

Client created!

Client created!

Client created!

Client created!

Experiment durations:11901.88, 3010.96, 3504.42, 2901.79, 2550.00

Experiment durations:12281.63, 3435.08, 2933.96, 2987.96, 3043.88

Experiment durations:12212.50, 3735.75, 2631.46, 2969.00, 2751.54

Experiment durations:12553.38, 4077.67, 2424.17, 2927.00, 3044.13

Experiment completed for file size: 128b

Running experiment for file size: **512b**

Client created!

Client created!

Client created!

Client created!

Experiment durations:11194.88, 4149.71, 3303.63, 4003.42, 3043.92

Experiment durations:11453.50, 4543.50, 3093.04, 4013.79, 3406.50

Experiment durations:11511.00, 4717.08, 2951.08, 4200.71, 3361.67

Experiment durations:11769.79, 4783.00, 3293.38, 4172.46, 3396.50

Experiment completed for file size: 512b

Running experiment for file size: **2kb**

Client created!

Client created!

Client created!

Client created!

Experiment durations:11589.88, 5459.38, 3855.92, 3127.25, 3102.21

Experiment durations:12164.00, 5321.38, 3739.58, 2797.92, 3017.67

Experiment durations:12311.25, 4655.96, 3503.17, 2693.71, 3144.75

Experiment durations:12784.00, 3671.29, 3362.29, 3376.08, 2125.00

Experiment completed for file size: 2kb

Running experiment for file size: **8kb**

Client created!

Client created!

Client created!

Client created!

Experiment durations:12755.04, 4340.50, 2807.67, 3477.79, 2631.00

Experiment durations:11431.71, 4267.92, 2952.79, 3372.67, 3006.67

Experiment durations:11211.88, 4373.13, 2796.54, 2928.38, 3047.67

Experiment durations:11341.92, 4503.33, 2902.88, 3763.29, 2843.75

Experiment completed for file size: 8kb

Running experiment for file size: **32kb**

Client created!

Client created!

Client created!

Client created!

Experiment durations:14009.38, 7039.25, 4266.92, 3864.54, 4248.38

Experiment durations:14324.04, 6775.00, 3968.33, 4213.58, 4082.46

Experiment durations:12532.38, 7009.58, 4132.29, 4036.46, 3907.75

Experiment durations:15336.00, 3751.83, 3857.71, 3727.38, 4628.08

Experiment completed for file size: 32kb